prob

1 Finite-Sample Coverage Guarantees for Mixed Boundary Sampling

We study a finite-horizon robust viable subset induced by a mixed boundary-supported control family. Let the discrete-time dynamics be

$$x_{k+1} = f_h(x_k, u_k), \qquad x_k \in X \subset \mathbb{R}^{2n}, \qquad u_k \in U := [-a_{\max}, a_{\max}]^n, \tag{1}$$

where $h > 0$ is the sampling time and $x = (q, \dot{q})$ denotes the robot state. Let $S \subset X$ denote a safe equilibrium set and define the shrunken safe set

$$X_\eta := \{x \in X : \operatorname{dist}(x, \partial X) \geq \eta\}, \qquad \eta > 0. \tag{2}$$

Mixed control family. Let ρ_{vert} be a distribution supported on the vertices

$$\operatorname{vert}(U) = \{-a_{\max}, +a_{\max}\}^n, \tag{3}$$

and let ρ_{sat} be a distribution supported on the ℓ_∞-boundary

$$\partial_\infty U := \{u \in U : \|u\|_\infty = a_{\max}\}, \tag{4}$$

with strictly positive density on $\partial_\infty U$. We define the mixed sampling distribution

$$\rho_{\text{mix}} := \lambda \rho_{\text{vert}} + (1 - \lambda)\rho_{\text{sat}}, \qquad \lambda \in (0, 1). \tag{5}$$

[Finite-horizon mixed viable target set] For $\eta > 0$ and horizon $K \in \mathbb{N}$, define

$$T^{\text{mix}}_{\eta,K} := \left\{ x_0 \in X_\eta : \begin{array}{l} \exists L \in \{1, \dots, K\},\ \exists u_0, \dots, u_{L-1} \in \operatorname{supp}(\rho_{\text{mix}}) \\ \text{s.t. } x_{k+1} = f_h(x_k, u_k), \\ x_k \in X_\eta,\ \forall k = 0, \dots, L, \\ x_L \in S \end{array} \right\}. \tag{6}$$

[Certified sample generator] A single certified sampling trial proceeds as follows:

1. sample a seed $s \sim \pi_S$ on S;
2. sample a horizon $L \sim \pi_L$ on $\{1, \dots, K\}$;
3. sample controls $u_0, \dots, u_{L-1} \overset{\text{i.i.d.}}{\sim} \rho_{\text{mix}}$;
4. construct a backward chain $x_L = s, x_{L-1}, \dots, x_0$ consistent with the dynamics;
5. accept the trial if every forward segment from x_k to x_{k+1} under u_k remains in X_η;

6. output one state chosen uniformly from the accepted chain.

Let ν_c denote the induced output distribution of accepted samples.

Lemma 1 (Soundness of certified samples)**.** *Every accepted output of the certified sample generator belongs to* $T^{\text{mix}}_{\eta,K}$*.*

Proof. Consider any accepted trial. By construction, the accepted chain admits a witness sequence of length $L \le K$ with controls in $\operatorname{supp}(\rho_{\text{mix}})$, all intermediate states remain in X_η, and the terminal state lies in S. Hence every state on the accepted chain belongs to $T^{\text{mix}}_{\eta,K}$. Since the output is selected from that chain, the claim follows. □

Assume that repeated trials are independently restarted, and let

$$X_1, \dots, X_N \overset{\text{i.i.d.}}{\sim} \nu_c \tag{7}$$

denote N certified samples.

Let μ_T denote the normalized volume measure on $T^{\text{mix}}_{\eta,K}$, namely

$$\mu_T(A) := \frac{\operatorname{vol}(A \cap T^{\text{mix}}_{\eta,K})}{\operatorname{vol}(T^{\text{mix}}_{\eta,K})}. \tag{8}$$

[ε-coverage fraction] For $\varepsilon > 0$, define the empirical coverage fraction

$$C_N(\varepsilon) := \mu_T\left(\bigcup_{i=1}^{N} B_\varepsilon(X_i)\right), \tag{9}$$

where $B_\varepsilon(x)$ is the closed ball of radius ε centered at x.

Assumption 1 (Uniform local mass lower bound)**.** There exists $p_\varepsilon > 0$ such that for all $x \in T^{\text{mix}}_{\eta,K}$,

$$\nu_c(B_\varepsilon(x)) \ge p_\varepsilon. \tag{10}$$

Theorem 1 (Finite-sample coverage guarantee)**.** *Under Assumption 1, the coverage fraction satisfies*

$$\mathbb{E}[C_N(\varepsilon)] \ge 1 - e^{-Np_\varepsilon}, \tag{11}$$

and for any $\gamma \in (0,1)$*,*

$$\mathbb{P}(C_N(\varepsilon) \ge \gamma) \ge 1 - \frac{e^{-Np_\varepsilon}}{1-\gamma}. \tag{12}$$

Consequently, for any $\delta \in (0,1)$*, if*

$$N \ge \frac{1}{p_\varepsilon} \log \frac{1}{\delta(1-\gamma)}, \tag{13}$$

then

$$\mathbb{P}(C_N(\varepsilon) \ge \gamma) \ge 1 - \delta. \tag{14}$$

Proof. Define the uncovered fraction

$$U_N(\varepsilon) := 1 - C_N(\varepsilon). \tag{15}$$

Fix any $y \in T^{\mathrm{mix}}_{\eta,K}$. The event that y is uncovered is equivalent to $X_i \notin B_\varepsilon(y)$ for all $i = 1, \dots, N$. Since the certified samples are i.i.d.,

$$\mathbb{P}(y \text{ is uncovered}) = \prod_{i=1}^{N} (1 - \nu_c(B_\varepsilon(y))) = (1 - \nu_c(B_\varepsilon(y)))^N. \tag{16}$$

By Assumption 1,

$$\mathbb{P}(y \text{ is uncovered}) \le (1 - p_\varepsilon)^N \le e^{-Np_\varepsilon}. \tag{17}$$

Integrating over y with respect to μ_T yields

$$\mathbb{E}[U_N(\varepsilon)] = \int_{T^{\mathrm{mix}}_{\eta,K}} \mathbb{P}(y \text{ is uncovered})\, d\mu_T(y) \le e^{-Np_\varepsilon}. \tag{18}$$

Therefore,

$$\mathbb{E}[C_N(\varepsilon)] = 1 - \mathbb{E}[U_N(\varepsilon)] \ge 1 - e^{-Np_\varepsilon}. \tag{19}$$

For the high-probability bound, by Markov's inequality,

$$\mathbb{P}(U_N(\varepsilon) > 1 - \gamma) \le \frac{\mathbb{E}[U_N(\varepsilon)]}{1 - \gamma} \le \frac{e^{-Np_\varepsilon}}{1 - \gamma}. \tag{20}$$

Since $U_N(\varepsilon) > 1 - \gamma$ is equivalent to $C_N(\varepsilon) < \gamma$, we obtain

$$\mathbb{P}(C_N(\varepsilon) \ge \gamma) \ge 1 - \frac{e^{-Np_\varepsilon}}{1 - \gamma}. \tag{21}$$

Setting the right-hand side to be at least $1 - \delta$ gives

$$e^{-Np_\varepsilon} \le \delta(1 - \gamma), \tag{22}$$

which is equivalent to

$$N \ge \frac{1}{p_\varepsilon} \log \frac{1}{\delta(1 - \gamma)}. \tag{23}$$

□

Assumption 2 (Density lower bound and geometric regularity)**.** Assume that ν_c admits a density g with respect to μ_T satisfying

$$g(x) \ge \alpha > 0, \qquad \forall x \in T^{\mathrm{mix}}_{\eta,K}, \tag{24}$$

and that $T^{\mathrm{mix}}_{\eta,K}$ is (c, d)-regular in the sense that

$$\mu_T(B_\varepsilon(x)) \ge c\,\varepsilon^d, \qquad \forall x \in T^{\mathrm{mix}}_{\eta,K}. \tag{25}$$

Corollary 1 (Explicit sample complexity)**.** *Under Assumption 2, Assumption 1 holds with*

$$p_\varepsilon = \alpha c \varepsilon^d. \tag{26}$$

Hence it suffices to take

$$N \ge \frac{1}{\alpha c \varepsilon^d} \log \frac{1}{\delta(1 - \gamma)} \tag{27}$$

to guarantee

$$\mathbb{P}(C_N(\varepsilon) \ge \gamma) \ge 1 - \delta. \tag{28}$$

Proof. For any $x \in T^{\mathrm{mix}}_{\eta,K}$,

$$\nu_c(B_\varepsilon(x)) = \int_{B_\varepsilon(x)\cap T^{\mathrm{mix}}_{\eta,K}} g(y)\, d\mu_T(y) \geq \alpha\mu_T(B_\varepsilon(x)) \geq \alpha c\varepsilon^d. \tag{29}$$

Thus Assumption 1 holds with $p_\varepsilon = \alpha c\varepsilon^d$, and the result follows from Theorem 1. $\square$

[Set inclusions for the three sampling families] Let $T^{\mathrm{vert}}_{\eta,K}$ and $T^{\mathrm{mix}}_{\eta,K}$ denote the target sets induced by the vertex-only and mixed control families, respectively. Then

$$T^{\mathrm{vert}}_{\eta,K} \subseteq T^{\mathrm{mix}}_{\eta,K}. \tag{30}$$

Moreover, if every admissible brake-only witness trajectory can be represented by a sequence in $\mathrm{supp}(\rho_{\mathrm{mix}})$, then

$$T^{\mathrm{brake}}_{\eta,K} \subseteq T^{\mathrm{mix}}_{\eta,K}. \tag{31}$$

Proof. The first inclusion follows directly from

$$\mathrm{supp}(\rho_{\mathrm{vert}}) \subseteq \mathrm{supp}(\rho_{\mathrm{mix}}). \tag{32}$$

The second inclusion follows from the representability assumption for brake-only witness sequences. $\square$